OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

13.8 Object marking

Marking describes the road marks of any objects like crosswalks, stopping-lines, and parking spaces. Marking is defined either in accordance to the bounding box of the element or by referencing outline points of the object.

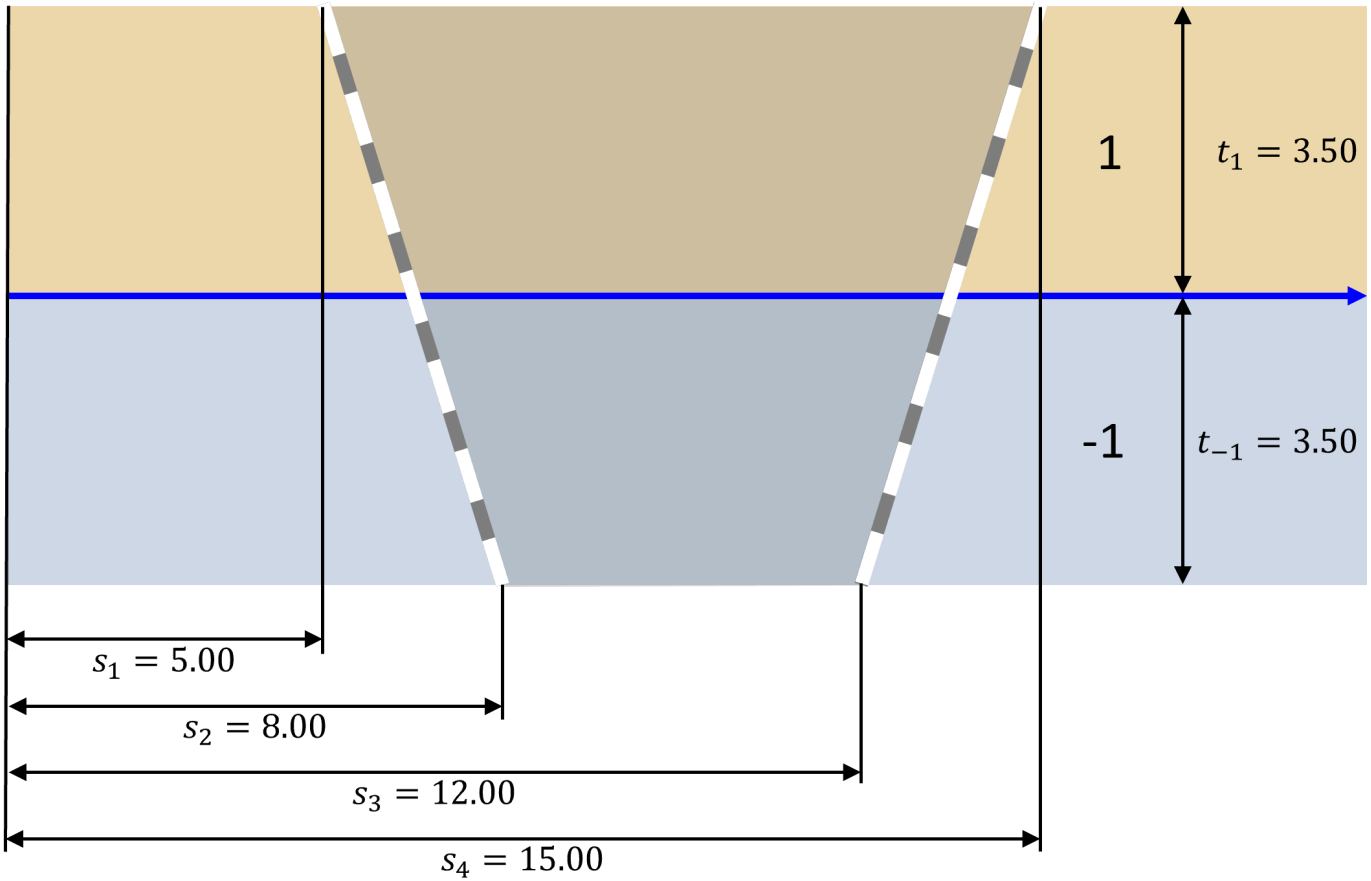

Figure 120. Crosswalk in ASAM OpenDRIVE

Figure 120 shows how a crosswalk with exemplary size is modeled.

The `<markings>` element serves as a wrapper for the `<marking>` element, which contains further information about the marking.

The marking may be defined for a straight line from one outline point to another by referencing the ID of the respective outline points. For this purpose, the `<cornerReference>` element inside the `<marking>` element is used.

Elements in UML model

`<markings>` element

In ASAM OpenDRIVE, the markings of objects are represented by the `<markings>` element within the `<object>` element.

```
UML class: t_road_objects_object_markings
XML tag:   <markings> (Multiplicity: 0..1)
```

Object markings are road markings of any objects, for example, crosswalks, stopping-lines, and parking spaces.

`<marking>` element

In ASAM OpenDRIVE, a single marking is represented by the `<marking>` element within the `<markings>` element.

```
UML class: t_road_objects_object_markings_marking
XML tag:   <marking> (Multiplicity: 1..*)
```

Specifies a marking that is either attached to one side of the object bounding box or referencing outline points.

Table 97. Attributes of the <marking> element

Name	Type	Use	Unit	Description
`color`	e_roadMarkColor	required		Color of the marking
`lineLength`	t_grZero	required	m	Length of the visible part

Name	Type	Use	Unit	Description
side	e_sideType	required		Side of the bounding box described in <object> element in the local coordinate system u/v
spaceLength	t_grEqZero	required	m	Length of the gap between the visible parts
startOffset	double	required	m	Lateral offset in u-direction from start of bounding box side where the first marking starts
stopOffset	double	required	m	Lateral offset in u-direction from end of bounding box side where the marking ends
weight	e_roadMarkWeight	optional		Optical "weight" of the marking
width	t_grZero	optional	m	Width of the marking
zOffset	t_grEqZero	optional	m	Height of road mark above the road, i.e. thickness of the road mark

<cornerReference> element

In ASAM OpenDRIVE, a corner reference is represented by the <cornerReference> element within the <marking> element.

```
UML class: t_road_objects_object_markings_marking_cornerReference
XML tag:   <cornerReference> (Multiplicity: 0..*)
```

Specifies a point by referencing an existing outline point.

Table 98. Attributes of the <cornerReference> element

Name	Type	Use	Description
id	nonNegativeInteger	required	Identifier of the referenced outline point

XML example

```
<objects>
    <object type="crosswalk"
            id="10"
            s="10.0"
            t="0.0"
            zOffset="0.0"
            orientation="none"
            length="10.0"
            width="7.0"
            hdg="0.0"
            pitch="0.0"
            roll="0.0">
        <outlines>
            <outline id="0">
                <cornerRoad s="5.0" t="3.5" dz="0.0" height="4.0" id="0"/>
                <cornerRoad s="8.0" t="-3.5" dz="0.0" height="4.0" id="1"/>
                <cornerRoad s="12.0" t="-3.5" dz="0.0" height="4.0" id="2"/>
                <cornerRoad s="15.0" t="3.5" dz="0.0" height="4.0" id="3"/>
            </outline>
        </outlines>
        <markings>
            <marking width="0.1"
                     color="white"
                     zOffset="0.005"
                     spaceLength ="0.05"
                     lineLength ="0.2"
                     startOffset="0.0"
                     stopOffset="0.0">
                <cornerReference id="0"/>
                <cornerReference id="1"/>
            </marking>
            <marking width="0.1"
                     color="white"
                     zOffset="0.005"
                     spaceLength ="0.05"
                     lineLength ="0.2"
```

```
                    startOffset="0.0"
                    stopOffset="0.0">
                <cornerReference id="2"/>
                <cornerReference id="3"/>
            </marking>
        </markings>
    </object>
</objects>
```

```
4
2
4
3
4
4
```

Rules

The following rules apply to object marking elements:

- The marking of an object shall either completely or partially be defined on its outline.
- asam.net:xodr:1.7.0:road.object.marking.colour: The color of the marking shall be defined.
- asam.net:xodr:1.7.0:road.object.marking.no_outline_side_attr: If no outline is used, the @side attribute is mandatory.
- asam.net:xodr:1.7.0:road.object.marking.outline_corner_reference_count: If an outline is used, at least two `<cornerReference>` elements are mandatory.
- asam.net:xodr:1.7.0:road.object.marking.no_cornerreference_if_no_outline: If no outline is used, the `<cornerReference>` element cannot be used.

Related topics

- Section 13.1, "Introduction to objects"
- Section 13.3, "Object outline"
- Section 13.9, "Object borders"